

Evading Local Maxima in ERC-7683 Intent Routing: A Max-SMT Approach

Abstract - As cross-chain intent networks mature, solver profitability is increasingly constrained by multi-dimensional inventory limits. Traditional fractional knapsack routing algorithms inherently underutilize solver capital by treating the mempool sequentially. This brief introduces Polytope, an SMT-based combinatorial allocator that increases aggregate yield by treating order flow as a globally evaluated matrix.

1. The Problem: Sequential Starvation

Current routing heuristics sort available intents by capital efficiency. While fast, this approach fails when constraints are non-linear or multi-dimensional. If a solver has \$5,000 USDC and \$3,000 WETH, a sequential algorithm might take a highly profitable USDC trade that exhausts the stablecoin cap. Because the algorithm evaluates linearly, it halts, leaving the \$3,000 WETH idle. The solver has hit a local maximum, leaving significant protocol revenue unextracted.

2. The Polytope Architecture

Polytope translates the solver's state into a Mixed-Integer Non-Linear Programming (MINLP) model using the Z3 Theorem Prover. The objective is to maximize the aggregate profit function.

This objective is strictly bound by simultaneous invariants:

- **Multi-Token Silos:** The sum of active payouts cannot exceed the inventory for any specific token.
- **Risk Boundaries:** Aggregate AMM slippage must not exceed predefined toxic thresholds.

3. Execution & Heuristic Seeding

To bypass the computational timeouts typical of SMT solvers, Polytope employs two mechanisms:

- **Heuristic Seeding (Warm Start):** The engine first runs a microsecond fractional knapsack sort. The resulting profit is injected into Z3 as an immutable lower bound, mathematically guaranteeing Polytope can never underperform the industry baseline.
- **Binary Search Iterative Tightening:** Rather than stepping linearly toward the global maximum, Polytope splits the profit search-space logarithmically, utilizing a 200ms execution guillotine to discard impossible branches rapidly.

4. Empirical Results

Across 15,000 historical Arbitrum intents, Polytope achieved a **+5.53% absolute profit edge** over the fractional knapsack. It accomplished this by dynamically unlocking and deploying

12.59% more idle capital (diverting flow to WETH and ARB) to maximize aggregate dollar return without breaching the stablecoin cap. With an average block optimization latency of ~355ms, Polytope operates well within the 1–5 second pre-flight windows demanded by modern intent auctions.